JAERYEONG KIM

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Legged locomotion, whole-body control, and reinforcement learning for quadruped and humanoid robots

EDUCATION

ETH Zurich September 2025 - Present
M.S. in Robotics, Systems, and Control *Zurich, Switzerland*

Korea Advanced Institute of Science and Technology March 2020 - August 2025
B.S. in Computer Science & Electrical and Electronic Engineering, Summa Cum Laude *Daejeon, South Korea*

ETH Zurich February 2024 - February 2025
ETH Zurich Exchange Program *Zurich, Switzerland*

- GPA 5.76/6.0 over 59 ECTS; semester project in deep reinforcement learning graded 6.0/6.0.

WORK EXPERIENCE

ETH Zurich — Computational Robotics Lab 2026 – Present
Master's Thesis — Motion Retargeting for Humanoid Loco-Manipulation *Zurich, Switzerland*

- Transferring human demonstrations that involve interactions with objects and terrain onto **humanoid robots** for loco-manipulation.

ETH Zurich — Computational Robotics Lab September 2025 – 2026
University Research Assistant — Dense Temporal Motion Retargeting (DTMR) *Zurich, Switzerland*

- Developed **DTMR**, which retargets human motion to legged robots by jointly optimizing timing and control with **sampling-based MPC**, so each robot follows the motion at a timing its own dynamics can execute.
- Integrated **MPPI** with GPU-parallelized MuJoCo simulation to generate kino-dynamically feasible motions from keypoint trajectories; paper in submission to **ICRA 2027**.

ETH Zurich — Robotic Systems Lab March 2026 – August 2026
Semester Project — Whole-Body Tactile Representations for Motion Control *Zurich, Switzerland*

- Benchmarked **whole-body tactile representations** for reinforcement learning-based dynamic locomotion, deployed on an **ANYmal-D** quadruped equipped with an FBG-based tactile skin.
- Designed a dodging benchmark task (tracking an end-effector target while evading balls launched at the shank) and compared proprioceptive baselines, an idealized single-point contact observation, and taxel-grid representations under one identical training recipe.
- Proposed **TaxelKV**, a multi-cell taxel-map representation matched to the real sensor; in simulation it cut impulse per contact from 1.30 to 0.45 N·s and end-effector tracking error from 0.027 to 0.011 m versus the proprioceptive baseline, and on hardware it remained robust to single-point localization noise where the baselines failed.

ETH Zurich — Computational Robotics Lab February 2024 - February 2025
University Research Assistant — Diversifying Gait Patterns using Unsupervised Skill Discovery *Zurich, Switzerland*

- Implemented **unsupervised skill discovery** methods with discrete latent variables (DIAYN, DISDAIN) and continuous latent variables (ASE, METRA) to encode skills into a latent space for control.
- Trained a policy that allows for controlling height, gait patterns, pitch angle, and feet width in quadruped locomotion.
- Deployed the RL-based multi-gait locomotion controller on real hardware for a live setting, used in the AATB artistic studio's interactive media art exhibition.

KAIST — Humanoid Generalization Lab June 2023 - August 2023
University Research Assistant — Hierarchical MCTS for Planning *Seoul, South Korea*

- Designed a hierarchical **Monte Carlo Tree Search** algorithm that searches all hierarchy levels from a single root, balancing exploration and exploitation to solve long-horizon tasks.
- Implemented the algorithm and evaluated it on a toy domain, a maze domain, and a **task and motion planning** domain.

STUDY PROJECTS

Real-to-Sim 6-DoF Object Pose Estimation Spring 2026

- Estimated 6-DoF object pose from a single egocentric RGB video stream (no depth sensor), tracking the object through a full manipulation sequence (grasping, carrying, dropping) to generate simulation-ready trajectories for **Isaac Sim**. 3D Vision course project at ETH Zurich.
- Implemented the dynamics-consistent trajectory replay and a **residual RL pipeline** that emits bounded corrections on the nominal robot configuration, jointly refining robot and object trajectories through simulated contact dynamics.

Visual Odometry Fall 2025

- Developed a monocular Visual Odometry pipeline for real-time camera pose estimation using KLT feature tracking, PnP-based localization, and landmark triangulation.
- Enhanced the baseline system with Local Bundle Adjustment, ground plane scale correction using SAM3 segmentation, and Sim(3)-based loop detection and closure, evaluated on the KITTI dataset.

TEACHING EXPERIENCE

ETH Zurich 2026 – Present
Teaching Assistant *Zurich, Switzerland*

- Planning and Decision Making for Robotics
- Linear System Theory

TECHNICAL STRENGTHS

Programming	Python, C/C++, PyTorch, ROS2
Simulation	NVIDIA Isaac Sim / Isaac Lab, MuJoCo
Methods	Sampling-based MPC, MPPI, Whole-Body Control, Motion Retargeting, Sim-to-Real, Unsupervised Skill Discovery
Robot Hardware	Unitree G02, Unitree G1, ANYmal, Booster K1
Languages	English (C1, TOEFL iBT 112/120), Korean (native)

HONORS

Scholarships

- Korea Presidential Science Scholarship *2020 - 2024*
- KAIST Presidential Fellowship (KPF), incl. funded UC Berkeley Summer Session (2022) *2020 - 2024*
- Korea Foundation for Advanced Studies (KFAS) Scholarship *2022 - 2023*

Awards

- Winner, Robot Competition, ETH Robotics Summer School (ETH Zurich RobotX): autonomous quadruped navigation stack (LiDAR-inertial estimation, FAR Planner). *2026*
- College of Engineering's Dean's List *Spring 2023*